Computational Economics

Lecture: Introduction to Markov Decision Processes

States, transitions, actions, and policies

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The Markov Process: Motivative Examples

- **Example: Days of the Week**

$$\mathcal{S} = \{\text{Mon}, \ldots, \text{Sun}\}.$$

Tomorrow follows deterministically from today. Starting on Monday, we return to Monday after 7, 14, 21, … days.
Idea: a Markov process can be deterministic and cyclical.

- **Example: An AR(1) Process**

$$s_{t+1} = \rho s_t + \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2).$$

Assume $\sigma > 0$; shocks are independent of the initial state. The current value determines the distribution of the next value.
Idea: states can be continuous. The process is Markov for any fixed ρ; $|\rho| < 1$ gives a stable AR(1) with a stationary distribution.

The Markov Process: Motivative Examples

Example: Employment and Unemployment

- Each month, an unemployed worker finds a job with probability α; an employed worker loses a job with probability δ, where $0 < \alpha, \delta < 1$.
- With states ordered as unemployed (U), employed (E),

$$P = \begin{pmatrix} 1-\alpha & \alpha \\ \delta & 1-\delta \end{pmatrix}.$$

  Rows give today's state; columns give next month's state. Each row sums to one.
- *Idea: transitions are random, even after we observe the state.* These probabilities are taken as given; the worker chooses no action in this model.
- How long does unemployment last? What fraction of time is spent unemployed in the long run?

Source: QuantEcon, Markov Chains: Basic Concepts.

The Markov Process: Motivative Examples

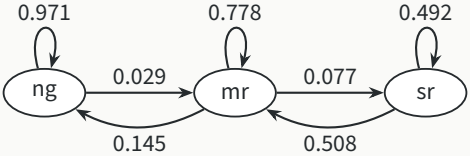
Example: U.S. Business Cycle Regimes

Monthly transition probabilities estimated by Hamilton (2005), as presented in QuantEcon:

$$P = \begin{pmatrix} 0.971 & 0.029 & 0 \\ 0.145 & 0.778 & 0.077 \\ 0 & 0.508 & 0.492 \end{pmatrix}.$$

States: normal growth (ng), mild recession (mr), severe recession (sr).

Idea: persistence differs across states. The probability of staying one more month is 97.1% in ng, 77.8% in mr, and 49.2% in sr.



The Markov Process: Formal Definition

Let $(s_t)_{t \geq 0}$ be a sequence of random variables taking values in $\mathcal{S}$.

- We say the sequence satisfies the **Markov property** if, for every t and every measurable subset $\bar{\mathcal{S}} \subseteq \mathcal{S}$,

$$\Pr(s_{t+1} \in \bar{\mathcal{S}} \mid s_0, \ldots, s_t) = \Pr(s_{t+1} \in \bar{\mathcal{S}} \mid s_t) = P_t(\bar{\mathcal{S}} \mid s_t),$$

  where the **transition kernel** $P_t(\bar{\mathcal{S}} \mid s)$ gives the probability of entering $\bar{\mathcal{S}}$ from state s at time t.

  Such a sequence is a **Markov process**: the current state contains all information from the history needed to predict the next state.

- We say a Markov process is **time-homogeneous** if $P_t = P$ for every t: its transition rule does not depend on calendar time.

- To specify its probability law, provide an **initial distribution** $s_0 \sim \mu_0$ and the transition kernels $(P_t)_{t \geq 0}$.

The Markov Process: Transition Properties

Consider a time-homogeneous Markov process on a finite state space.

- Its transition kernel is represented by a **transition matrix** P:

$$P_{ij} = \Pr(s_{t+1} = j \mid s_t = i), \qquad P_{ij} \geq 0, \qquad \sum_j P_{ij} = 1.$$

  A matrix with nonnegative entries and unit row sums is called **row-stochastic**.

- If μ_t is a row vector of state probabilities,

$$\mu_{t+1} = \mu_t P, \qquad \mu_{t+n} = \mu_t P^n.$$

- The probability of moving from i to j in n steps is $(P^n)_{ij}$. Summing over intermediate states gives $P^{n+m} = P^n P^m$.

- **Conditional expectation:** for a column vector g of state-dependent outcomes,

$$\mathbb{E}[g(s_{t+1}) \mid s_t = i] = \sum_j P_{ij} g(j) = (Pg)_i.$$

The Markov Process: Reachability and Periodicity

Consider a finite, time-homogeneous chain.

- We say the chain is **irreducible** if every state can reach every other state:

$$\text{for every } i, j, \quad \exists n \geq 0 \text{ such that } (P^n)_{ij} > 0.$$

  Positive one-step probabilities between every pair are unnecessary; a path through other states is enough.

- For a state i with a possible return, define its **period** by

$$d(i) = \gcd\{n \geq 1 : (P^n)_{ii} > 0\}.$$

  An irreducible chain has a common period. We call it **aperiodic** if this period is 1, and **periodic** otherwise.

- The days-of-the-week chain is irreducible with period 7. The employment chain is irreducible and aperiodic: both states communicate and have self-loops.

The Markov Process: Stationarity and Long-run Behavior

For a finite, time-homogeneous chain:

- A probability row vector π is a **stationary distribution** of the chain if

$$\pi P = \pi, \qquad \pi_i \geq 0, \qquad \sum_i \pi_i = 1.$$

  Every finite chain has at least one. If $\mu_0 = \pi$, every $\mu_t = \pi$.

- If the chain is **irreducible**, its stationary distribution is unique and

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbf{1}\{s_t = i\} \longrightarrow \pi_i \quad \text{almost surely.}$$

- If the chain is also **aperiodic**, then for every initial distribution μ_0,

$$\mu_0 P^n \longrightarrow \pi.$$

  The weekly cycle has uniform visit frequencies but its distribution keeps cycling when started on Monday.

The Markov Decision Process: Motivative Examples

Example: Maze Escape

- **State:** current free cell. **Action:** up, down, left, or right.
- **Transition:** move one cell; hitting a wall or boundary leaves the position unchanged. The goal G is absorbing.
- **Reward:** -1 per move while outside G; zero thereafter. Choose actions to reach G quickly, with discount factor $0 < \beta < 1$.

Idea: actions control the next state; rewards express the goal.

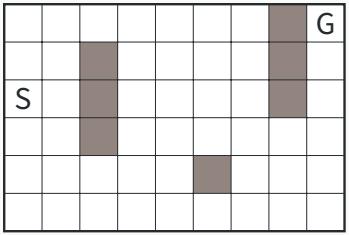


Figure 1: Maze navigation.

The Markov Decision Process: Motivative Examples

Example: McCall Job Search

- **State:** $\mathrm{U}(w)$, unemployed with current wage offer w, or $\mathrm{E}(w)$, employed at wage w.
- **Action and reward:** at $\mathrm{U}(w)$, accept and earn w today, or reject and receive benefit b today.
- **Transition:** rejection brings a new independent offer $w' \sim F$. Acceptance leads to $\mathrm{E}(w)$, with wage w every period and no job separation. At $\mathrm{E}(w)$, the only action is to stay employed.
- **Objective:** maximize expected discounted income, $0 < \beta < 1$. Assume wages have bounded support and b is finite.
- *Idea: a discrete choice changes future opportunities.* Waiting preserves the chance of a better offer; accepting locks in income.

Source: QuantEcon, The McCall Search Model.

The Markov Decision Process: Motivative Examples

Example: Cake Eating

- **State:** remaining stock $k_t \in [0, \bar{k}]$, where $\bar{k} < \infty$.
- **Action:** consume $c_t \in [0, k_t]$.
- **Transition:** $k_{t+1} = k_t - c_t$; the unconsumed stock is saved. There is no production or uncertainty.
- **Reward:** $u(c_t) = \sqrt{c_t}$, including $u(0) = 0$.
- **Objective:** choose consumption over time to maximize

$$\sum_{t=0}^{\infty} \beta^t \sqrt{c_t}, \qquad 0 < \beta < 1.$$

- *Idea: both states and actions can be continuous, and feasible actions depend on the state.* Consuming more today leaves less for tomorrow.

The Markov Decision Process: Formal Definition

We focus on a fully observed, time-homogeneous, infinite-horizon discounted MDP:

$$(\mathcal{S}, \{\mathcal{A}(s)\}_{s\in\mathcal{S}}, P, r, \beta), \qquad 0 < \beta < 1.$$

- **States and actions:** observe $s_t \in \mathcal{S}$, then choose $a_t \in \mathcal{A}(s_t)$, a nonempty feasible set.
- The state dynamics satisfy the **controlled Markov property** if, for every t and measurable $\bar{\mathcal{S}} \subseteq \mathcal{S}$,

  $$\Pr(s_{t+1} \in \bar{\mathcal{S}} \mid s_0, a_0, \ldots, s_t, a_t) = P(\bar{\mathcal{S}} \mid s_t, a_t).$$

  where P is the time-independent **controlled transition kernel**.
- The **reward function** r specifies expected payoffs: receive R_{t+1} after action a_t, with conditional mean $r(s_t, a_t)$ given the history and current action. Rewards and next states may be jointly random.
- The **discount factor** β weights future rewards. Time homogeneity means that both P and r are independent of t.

The Markov Decision Process: Timing

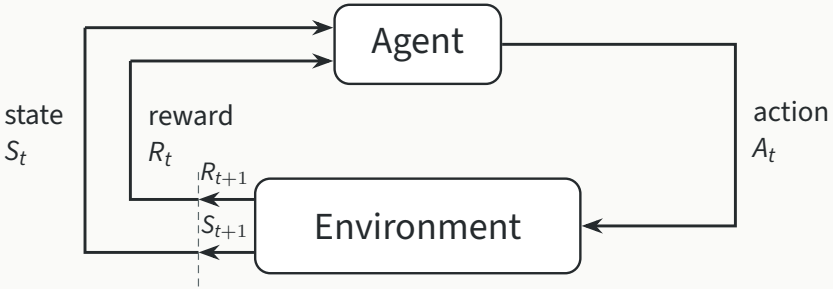


Figure 2: Agent–environment loop (adapted from Sutton and Barto (2018), Fig. 3.1).

Observe s_t, choose a_t, then receive R_{t+1} and observe s_{t+1}.
$r(s_t, a_t)$ is the expected reward from that action.

The Markov Decision Process: Policies

- A policy specifies how actions are chosen. It is a **Markov policy** if its action distribution depends only on the current state and time.
- A Markov policy is **stationary** if it does not depend on time, and **deterministic** if it selects one action with probability one. Such a policy is a feasible rule

$$a_t = \sigma(s_t), \qquad \sigma(s) \in \mathcal{A}(s).$$

A stationary randomized policy instead specifies a distribution $\sigma(\cdot \mid s)$ over feasible actions.

The Markov Decision Process: A Policy Induces a Chain

For a finite state space, fix a stationary deterministic policy σ.

- The induced transition matrix and expected reward vector are

$$P_{ij}^{\sigma} = P(j \mid i, \sigma(i)), \qquad r_i^{\sigma} = r(i, \sigma(i)).$$

- P^{σ} is row-stochastic: fixing the decision rule gives a time-homogeneous Markov chain. Thus $\mu_{t+1} = \mu_t P^{\sigma}$.
- Different policies can change reachability and long-run behavior. An absorbing goal or accepted job can make P^{σ} reducible; the earlier convergence conditions must be checked for each policy.

References and Further Reading

1. **Richard S. Sutton and Andrew G. Barto (2018), Reinforcement Learning: An Introduction. 2nd ed. MIT Press. [Book]**
   - Chapter 3: Finite Markov Decision Processes
2. **Dimitri P. Bertsekas (2019), Reinforcement Learning and Optimal Control. Athena Scientific. [Book]**
   - Chapter 1: Exact Dynamic Programming (selected sections)
3. **Dimitri P. Bertsekas and Yuchao Li (2025), Topics in Reinforcement Learning. CSE 691, Arizona State University. [Online course]**
   - Lecture 1: Introduction to Sequential Decision Making (Reinforcement Learning, Dynamic Programming, Decision and Control)
   - Lecture 2: Stochastic Finite and Infinite Horizon DP, Approximation in Value Space

References and Further Reading

4. Thomas J. Sargent and John Stachurski, QuantEcon. [Online lectures]

- Markov Chains: Basic Concepts
- Markov Chains: Irreducibility and Ergodicity
- Job Search I: The McCall Search Model
- Discrete State Dynamic Programming